ANALYSTLAB AFRICA

Experience Lab Internship Programme

Official Weekly Assignment

Week 5: SwiftLog Logistics Analytics & AI Transformation Project

Document Control

Programme	AnalystLab Africa Experience Lab
Week	Week 5
Project	SwiftLog Logistics Analytics & AI Transformation Project
Project Title	Enterprise Supply Chain Analytics and Artificial Intelligence Transformation for SwiftLog Logistics Limited
Version	1.0
Issue Date	_____
Submission Deadline	End of Week 5 (Sunday, 11:59 PM WAT)
Submission Method	Official AnalystLab Africa Week 5 Submission Form

1. DATA ANALYTICS TRACK

Project Title

SwiftLog Logistics Supply Chain Analytics & Business Intelligence

Project Overview

Congratulations!

You have recently joined **AnalystLab Africa Consulting** as a **Junior Data Analyst**.

Your consulting team has been assigned to work with **SwiftLog Logistics Limited**, one of Nigeria's rapidly growing logistics and supply chain companies.

SwiftLog operates warehouses, suppliers, inventory systems, shipment tracking, customer delivery services, and nationwide logistics operations. Due to rapid expansion, the company is experiencing

operational inefficiencies, delayed deliveries, inventory challenges, supplier performance issues, and limited executive visibility into key business metrics.

Your team has been hired to design a comprehensive Business Intelligence solution that enables management to monitor business performance, identify operational bottlenecks, and make data-driven decisions.

As the assigned Data Analyst, your role is to transform raw logistics data into meaningful insights through data cleaning, modeling, visualization, dashboard development, and executive reporting.

Business Objectives

Your solution should help SwiftLog management:

- Monitor business performance.
- Improve delivery efficiency.
- Reduce operational costs.
- Improve inventory management.
- Evaluate supplier performance.
- Improve warehouse operations.
- Reduce product returns.
- Support executive decision-making.

Datasets to Use

For this project, you are required to use the following datasets provided as part of the **SwiftLog Logistics Analytics & AI Transformation Project**. You are expected to integrate multiple datasets where necessary to perform comprehensive business analysis, develop KPIs, and build an interactive Power BI dashboard.

Dataset	Purpose
SwiftLog_Customers.csv	Analyse customer demographics, customer distribution, purchasing behaviour, customer order frequency, and identify high-value customers.
SwiftLog_Orders.csv	Analyse order trends, sales performance, order status, monthly order volumes, and customer purchasing patterns.
SwiftLog_Order_Items.csv	Analyse product sales, quantities ordered, revenue contribution, and identify top-selling and low-performing products.
SwiftLog_Products.csv	Evaluate product performance, product categories, pricing, and profitability across different product groups.

Dataset	Purpose
SwiftLog_Suppliers.csv	Analyse supplier performance, supplier distribution, and identify suppliers contributing to operational efficiency or delays.
SwiftLog_Warehouses.csv	Analyse warehouse locations, warehouse operations, regional performance, and warehouse capacity utilisation.
SwiftLog_Inventory.csv	Monitor inventory levels, identify low-stock products, analyse inventory turnover, and evaluate stock availability across warehouses.
SwiftLog_Shipments.csv	Analyse shipment trends, shipment status, warehouse dispatch activities, and logistics operations.
SwiftLog_Delivery_Performance.csv	Evaluate delivery performance, identify delayed shipments, calculate on-time delivery rates, and analyse logistics efficiency.
SwiftLog_Employees.csv	Analyse workforce distribution across warehouses and evaluate staffing levels in relation to warehouse operations.
SwiftLog_Returns.csv	Analyse product returns, return reasons, return rates, and identify products requiring quality or logistics improvements.
SwiftLog_Data_Dictionary.xlsx	Use as a reference to understand dataset fields, business definitions, and relationships between datasets.

Dataset Integration Requirements

To answer the business questions effectively, you are expected to integrate multiple datasets where appropriate. At a minimum, your analysis should include the following relationships:

Business Analysis	Recommended Datasets
Customer Performance Analysis	Customers + Orders
Product Sales & Revenue Analysis	Orders + Order_Items + Products
Customer Purchasing Behaviour	Customers + Orders + Order_Items
Inventory & Warehouse Analysis	Inventory + Products + Warehouses
Shipment & Delivery Performance	Orders + Shipments + Delivery_Performance
Supplier Performance Analysis	Suppliers + Products + Shipments

Business Analysis	Recommended Datasets
Product Return Analysis	Orders + Order_Items + Products + Returns
Executive KPI Dashboard	Integration of all relevant datasets

Note: You are encouraged to create additional relationships where necessary to answer business questions and produce meaningful insights. Your Power BI data model should be well-structured, efficient, and support interactive reporting.

Project Tasks

Part 1 — Data Import, Understanding & Preparation (15 Marks)

Import all datasets into Power BI.

Perform the following:

- Inspect every dataset.
- Verify column names and data types.
- Identify primary and foreign keys.
- Handle missing values.
- Remove duplicate records.
- Correct inconsistent records.
- Rename fields where necessary.
- Build appropriate table relationships.
- Develop a professional star schema or relational data model.
- Create a data dictionary for any additional calculated fields you introduce.
- Capture screenshots of your completed data model.

Part 2 — Data Cleaning & Transformation (20 Marks)

Using Power Query:

- Clean all datasets.
- Remove unnecessary columns.
- Standardize text values.
- Split and merge columns where appropriate.
- Create Date, Month, Quarter and Year columns.
- Create conditional columns.
- Create calculated columns.
- Create custom business fields.
- Optimize the data model for performance.
- Document every transformation performed.

Part 3 — KPI Development & DAX Measures (25 Marks)

Develop meaningful KPIs using DAX.

Your report should include, at a minimum:

- Total Revenue
- Total Orders
- Total Customers
- Total Products
- Total Shipments
- Average Order Value
- Average Delivery Time
- Average Delivery Delay
- Inventory Stock Level
- Inventory Turnover
- Return Rate
- Supplier Performance Score
- On-Time Delivery Rate
- Warehouse Utilization
- Customer Retention Indicator (or another relevant customer metric)

Create **at least 10 custom DAX measures** and explain their business value.

Part 4 — Dashboard Development (60 Marks)

Develop a professional Power BI solution consisting of **at least five interactive report pages**.

Executive Dashboard

Include:

- KPI Cards
- Revenue Trend
- Monthly Orders
- Delivery Performance
- Return Overview
- Executive Filters

Customer Analytics Dashboard

Include:

- Top Customers
- Customer Revenue

- Customer Order Frequency
- Geographic Distribution
- Customer Growth Trend

Inventory & Warehouse Dashboard

Include:

- Inventory Levels
- Warehouse Capacity
- Inventory Turnover
- Low Stock Products
- Warehouse Comparison

Shipment & Delivery Dashboard

Include:

- Delivery Status
- Delayed Deliveries
- Delivery Performance by Region
- Average Delivery Time
- Shipment Trend

Supplier & Product Dashboard

Include:

- Supplier Performance
- Product Categories
- Best Selling Products
- Returned Products
- Revenue by Product

Your Power BI solution **must also include**:

- Drill-through pages
- Bookmarks
- Dynamic tooltips
- Report page navigation buttons
- Slicers and synchronized filters
- Cross-filtering visuals
- Dynamic report titles
- Conditional formatting
- Advanced DAX measures
- What-If Parameters (where appropriate)

- Mobile report layout
- Professional colour theme and branding

Part 5 — Business Analysis & Data Storytelling (25 Marks)

Analyse the dashboards and answer the following questions:

1. Which products generate the highest revenue?
2. Which customers contribute the highest revenue?
3. Which warehouses operate most efficiently?
4. Which suppliers consistently perform well?
5. Which products have the highest return rates?
6. Which regions experience the most delayed deliveries?
7. Which months record peak business activity?
8. Which KPIs require immediate management attention?
9. What operational inefficiencies were identified?
10. What business opportunities should SwiftLog pursue?

Support every answer with dashboard visuals and business evidence.

Part 6 — Business Recommendations (20 Marks)

Provide **at least ten strategic recommendations**.

Recommendations should address:

- Delivery optimization
- Inventory management
- Warehouse efficiency
- Supplier improvement
- Customer retention
- Operational cost reduction
- Logistics optimization
- Return reduction
- Performance monitoring
- Business growth opportunities

Every recommendation must be supported by findings from your dashboard.

Part 7 — Executive Reporting & Presentation (15 Marks)

Prepare a professional executive presentation (10–15 slides).

Your presentation should include:

- Business Overview
- Project Objectives
- Data Preparation Process
- Dashboard Demonstration
- KPI Summary
- Key Insights
- Business Recommendations
- Expected Business Impact
- Conclusion

In addition, prepare a **3–5 page Executive Business Intelligence Report** summarizing your findings for SwiftLog's leadership team.

Expected Deliverables

Submit your work in the following structure:

```
Week5_DataAnalytics_SwiftLog_YourName/

├── Data/
│   └── Original Datasets
├── Dashboard/
│   └── SwiftLog_Analytics.pbix
├── Report/
│   ├── Executive_BI_Report.pdf
│   └── Dashboard_Screenshots.pdf
├── Presentation/
│   └── Executive_Presentation.pptx
├── Images/
├── README.md
└── References/
```

PART 10 — ASSESSMENT CRITERIA

Section	Marks
Data Preparation & Modeling	15
Data Cleaning & Transformation	20
KPI Development & DAX	25
Dashboard Development	60
Business Analysis & Storytelling	25
Business Recommendations	20
Executive Report & Presentation	15
Documentation & Project Organization	20
Total	**200 Marks**

Expected Learning Outcomes

By the end of this project, interns should be able to:

- Build a complete end-to-end Business Intelligence solution using multiple related datasets.
- Apply advanced Power Query transformations and data modeling techniques.
- Develop business-focused KPIs and DAX measures.
- Design interactive, executive-ready Power BI dashboards with advanced navigation and interactivity.
- Translate analytical findings into actionable business recommendations.
- Present insights professionally through reports and executive presentations.

2. DATA SCIENCE TRACK

Mission DS05: Supply Chain Intelligence & Feature Engineering for Predictive Logistics

Client

SwiftLog Logistics Limited

SwiftLog Logistics Limited is a rapidly growing logistics and supply chain company operating across multiple regions. The company manages customer orders, warehouses, inventory, suppliers, shipments, deliveries, employees, and product returns.

As the business continues to expand, management has observed increasing delivery delays, inventory imbalances, supplier performance inconsistencies, and operational bottlenecks. Before investing in predictive analytics and Artificial Intelligence solutions, the executive team wants to gain a deeper understanding of its operational data.

As a **Junior Data Scientist** at **AnalystLab Africa Consulting**, your responsibility is to integrate, clean, and analyse the company's logistics datasets to uncover hidden patterns, identify operational inefficiencies, engineer meaningful features, and generate actionable business insights. Your work will produce a high-quality analytical dataset that will serve as the foundation for future Machine Learning and AI solutions.

Learning Objectives

By completing this project, you should be able to:

- Acquire and integrate multiple datasets into a unified analytical dataset.
- Perform comprehensive data cleaning and preprocessing.
- Conduct exploratory data analysis (EDA).
- Apply descriptive and inferential statistical techniques.
- Engineer meaningful features for predictive modelling.
- Identify hidden patterns and relationships within logistics operations.
- Segment customers, suppliers, or warehouses based on business characteristics.
- Prepare high-quality datasets for Machine Learning.
- Communicate insights through professional data storytelling and reporting.

Datasets to Use

For this project, you are required to use the following datasets provided as part of the **SwiftLog Logistics Analytics & AI Transformation Project**. You are expected to integrate multiple datasets to perform comprehensive exploratory data analysis, statistical analysis, feature engineering, and business insight generation.

Dataset	Purpose
SwiftLog_Customers.csv	Analyse customer behaviour, customer segmentation, customer order frequency, and estimate customer value.
SwiftLog_Orders.csv	Analyse order trends, customer purchasing behaviour, operational performance, and order patterns.
SwiftLog_Order_Items.csv	Analyse product-level transactions, quantities sold, basket composition, and support feature engineering.
SwiftLog_Products.csv	Evaluate product performance, product categories, pricing patterns, and profitability.
SwiftLog_Suppliers.csv	Analyse supplier reliability, supplier performance, and supplier contribution to logistics operations.
SwiftLog_Warehouses.csv	Evaluate warehouse operations, regional distribution, and warehouse performance.
SwiftLog_Inventory.csv	Analyse inventory levels, stock movement, inventory turnover, and replenishment needs.
SwiftLog_Shipments.csv	Analyse shipment operations, shipment patterns, logistics flow, and transportation activities.
SwiftLog_Delivery_Performance.csv	Evaluate delivery delays, delivery efficiency, and operational performance for predictive analysis.
SwiftLog_Employees.csv	Analyse workforce distribution and warehouse staffing patterns.
SwiftLog_Returns.csv	Analyse return trends, return reasons, and product quality issues.
SwiftLog_Data_Dictionary.xlsx	Reference guide for understanding dataset fields, business definitions, and relationships between datasets.

Dataset Integration Requirements

To complete this project successfully, you are expected to integrate multiple datasets where appropriate.

At a minimum, your analysis should include the following relationships:

Analysis	Recommended Datasets
Customer Behaviour Analysis	Customers + Orders
Product Performance Analysis	Orders + Order_Items + Products
Customer Purchase Analysis	Customers + Orders + Order_Items
Inventory Analysis	Inventory + Products + Warehouses
Delivery Performance Analysis	Orders + Shipments + Delivery_Performance
Supplier Performance Analysis	Suppliers + Products + Shipments
Product Return Analysis	Orders + Order_Items + Products + Returns
Workforce Analysis	Employees + Warehouses
Feature Engineering	Integration of all relevant datasets

Note: You are encouraged to create additional relationships where necessary. Your final analytical dataset should be clean, well-structured, and suitable for Machine Learning tasks.

Project Tasks

Part 1 — Data Acquisition & Understanding (15 Marks)

Using Python, import all required datasets into Jupyter Notebook.

Perform the following:

- Import all datasets.
- Inspect dataset structure.
- Identify data types.
- Determine the number of rows and columns.
- Identify primary and foreign keys.
- Produce summary statistics.
- Assess data quality.

- Create a data relationship diagram showing how datasets are connected.

Part 2 — Data Cleaning & Preprocessing (25 Marks)

Prepare the datasets for analysis by:

- Handling missing values.
- Removing duplicate records.
- Correcting inconsistent records.
- Standardising categorical values.
- Converting appropriate data types.
- Detecting and treating outliers.
- Encoding categorical variables where appropriate.
- Creating reusable preprocessing functions or pipelines.
- Generating an automated data profiling report using **ydata-profiling**, **Sweetviz**, or a similar tool.
- Documenting every preprocessing step.

Part 3 — Exploratory Data Analysis (35 Marks)

Perform a comprehensive Exploratory Data Analysis (EDA).

Your analysis should include:

Univariate Analysis

- Customers
- Orders
- Products
- Inventory
- Shipments
- Returns
- Suppliers
- Warehouses

Bivariate Analysis

- Customer vs Orders
- Product Category vs Revenue
- Warehouse vs Inventory Levels
- Supplier vs Delivery Performance
- Delivery Delay vs Returns
- Region vs Shipment Performance

Multivariate Analysis

Investigate relationships among multiple variables affecting logistics performance.

Create **at least 15 professional visualisations** using Matplotlib, Seaborn, or Plotly.

Each visualisation must include:

- Title
- Labels
- Interpretation
- Business implication

Part 4 — Statistical Analysis (20 Marks)

Perform descriptive and inferential statistical analysis.

Include:

- Mean
- Median
- Mode
- Variance
- Standard Deviation
- Correlation Analysis
- Distribution Analysis
- Skewness
- Kurtosis

Conduct **at least three hypothesis tests** relevant to the business, such as:

- Delivery performance across warehouses.
- Supplier performance comparison.
- Product return rates across categories.
- Delivery delays across regions.

Interpret all statistical results from a business perspective.

Part 5 — Feature Engineering (30 Marks)

Create meaningful features that will improve predictive modelling.

Include features such as:

- Delivery Duration
- Delivery Delay Flag
- Customer Lifetime Value (estimated)
- Average Customer Spend
- Order Frequency
- Inventory Turnover
- Supplier Reliability Score

- Warehouse Utilisation Rate
- Return Frequency
- Delivery Efficiency Score
- Seasonal Demand Indicator
- Monthly Demand Trend

Explain:

- Why each feature was created.
- Its business value.
- How it could improve future Machine Learning models.

Part 6 — Segmentation & Pattern Discovery (20 Marks)

Perform exploratory segmentation using suitable analytical techniques.

Examples include:

- Customer Segmentation
- Product Segmentation
- Supplier Segmentation
- Warehouse Segmentation
- Regional Segmentation

Visualise each segment and explain how SwiftLog could use the findings to improve business performance.

Part 7 — Business Insights & Recommendations (25 Marks)

Answer the following questions:

1. Which customer segments generate the highest business value?
2. Which products contribute most to revenue?
3. Which suppliers consistently underperform?
4. Which warehouses require operational improvements?
5. Which products have the highest return rates?
6. Which regions experience the greatest delivery delays?
7. What operational bottlenecks have you identified?
8. Which variables are most suitable for predictive modelling?
9. What hidden patterns did your analysis uncover?
10. What recommendations would you make to improve SwiftLog's operations?

Support every answer with evidence from your analysis.

Provide **at least ten strategic recommendations** based on your findings.

Part 8 — Documentation & Presentation (15 Marks)

Prepare a professional submission including:

Technical Report

- Executive Summary
- Data Understanding
- Data Cleaning Process
- Exploratory Data Analysis
- Statistical Analysis
- Feature Engineering
- Segmentation Results
- Business Insights
- Recommendations
- Conclusion

Executive Presentation

Prepare a **10–15 slide presentation** summarising:

- Business Problem
- Data Science Methodology
- Key Findings
- Visualisations
- Recommendations
- Future Machine Learning Opportunities

Expected Deliverables

Submit your work in the following structure:

```
Week5_DataScience_SwiftLog_YourName/

├── Data/
│   ├── Original_Datasets/
│   ├── Cleaned_Datasets/
│   └── Feature_Engineered_Dataset.csv
│
├── Notebook/
│   └── Week5_DataScience.ipynb
│
├── Reports/
│   ├── Data_Profiling_Report.html
│   ├── Technical_Report.pdf
│   └── Executive_Summary.pdf
│
├── Visualizations/
│   ├── Charts/
│   └── Dashboards/
│
├── Presentation/
│   └── Executive_Presentation.pptx
```

|

PART 9 — ASSESSMENT CRITERIA

Section	Marks
Data Acquisition & Understanding	15
Data Cleaning & Preprocessing	25
Exploratory Data Analysis	35
Statistical Analysis	20
Feature Engineering	30
Segmentation & Pattern Discovery	20
Business Insights & Recommendations	25
Technical Report & Presentation	15
Documentation & Project Organisation	15
Total	**200 Marks**

Machine Learning Track

Week 5 – Logistics Machine Learning

Mission ML05: Intelligent Logistics Prediction & Supply Chain Optimization

Client

SwiftLog Logistics Limited

SwiftLog Logistics Limited is a rapidly growing logistics and supply chain company operating across multiple regions. The company manages customer orders, warehouses, suppliers, inventory, shipments, deliveries, employees, and product returns.

Despite significant business growth, SwiftLog faces challenges including delayed deliveries, inventory shortages, supplier inconsistencies, increasing product returns, and rising operational costs. Management wants to move beyond descriptive analytics and begin leveraging **Machine Learning** to make proactive, data-driven decisions.

As a **Junior Machine Learning Engineer** at **AnalystLab Africa Consulting**, your responsibility is to design, develop, evaluate, and deploy predictive models that help SwiftLog improve operational

efficiency, reduce delays, optimize inventory management, and support strategic decision-making. Your models should demonstrate how Machine Learning can create measurable business value across the organization.

Business Objectives

Your Machine Learning solutions should help SwiftLog management:

- Predict delayed deliveries before they occur.
- Forecast product demand and inventory requirements.
- Identify customers likely to return products.
- Predict supplier reliability.
- Improve warehouse operations.
- Reduce logistics costs.
- Improve customer satisfaction.
- Support AI-driven decision making.

Datasets to Use

Use the following datasets to complete this project.

Dataset	Purpose
SwiftLog_Customers.csv	Customer behaviour analysis, customer profiling and predictive customer insights.
SwiftLog_Orders.csv	Order trends, order history and operational performance.
SwiftLog_Order_Items.csv	Product-level transactions for predictive modelling and feature engineering.
SwiftLog_Products.csv	Product information, product categories and pricing.
SwiftLog_Suppliers.csv	Supplier information and supplier reliability modelling.
SwiftLog_Warehouses.csv	Warehouse operations and warehouse performance modelling.
SwiftLog_Inventory.csv	Inventory levels, stock movement and inventory forecasting.
SwiftLog_Shipments.csv	Shipment tracking, logistics activities and delivery prediction.
SwiftLog_Delivery_Performance.csv	Delivery performance data used for delay prediction and operational optimization.
SwiftLog_Employees.csv	Workforce allocation and operational resource analysis.

Dataset	Purpose
SwiftLog_Returns.csv	Product return prediction and customer satisfaction analysis.
SwiftLog_Data_Dictionary.xlsx	Reference guide for dataset fields and business definitions.

Dataset Integration Requirements

At a minimum, your models should integrate the following datasets:

Machine Learning Problem	Recommended Datasets
Delivery Delay Prediction	Orders + Shipments + Delivery_Performance
Demand Forecasting	Orders + Order_Items + Products + Inventory
Product Return Prediction	Orders + Order_Items + Customers + Returns
Supplier Reliability Prediction	Suppliers + Shipments + Delivery_Performance
Customer Value Prediction	Customers + Orders + Order_Items
Warehouse Performance Prediction	Warehouses + Inventory + Shipments

Learning Objectives

By completing this project, you should be able to:

- Prepare datasets for Machine Learning.
- Engineer meaningful predictive features.
- Train supervised learning models.
- Compare multiple Machine Learning algorithms.
- Tune model hyperparameters.
- Evaluate model performance using appropriate metrics.
- Interpret feature importance.
- Explain model predictions.
- Deploy a basic Machine Learning application.
- Communicate Machine Learning insights to business stakeholders.

Project Tasks

Part 1 — Data Preparation & Feature Engineering (20 Marks)

Using the provided datasets:

- Import all required datasets.
- Clean and preprocess the data.
- Handle missing values.
- Encode categorical variables.
- Scale or normalize numerical variables where appropriate.
- Engineer meaningful predictive features.
- Split the data into training, validation and testing datasets.
- Document your preprocessing workflow.

Part 2 — Exploratory Data Analysis for Machine Learning (20 Marks)

Perform exploratory analysis to understand relationships between predictor variables and target variables.

Include:

- Correlation analysis
- Feature distributions
- Target variable distribution
- Class imbalance analysis
- Missing value analysis
- Outlier detection
- Feature importance exploration

Support your analysis with appropriate visualizations.

Part 3 — Build Predictive Models (50 Marks)

Develop **at least four Machine Learning models** addressing different logistics problems.

Examples include:

- Delivery Delay Prediction (Classification)
- Product Return Prediction (Classification)
- Inventory Demand Forecasting (Regression)
- Supplier Reliability Prediction (Classification)
- Customer Value Prediction (Regression)
- Warehouse Performance Prediction (Regression or Classification)

For each problem:

- Select an appropriate target variable.
- Justify the choice of features.
- Train at least **three different algorithms**.
- Compare model performance.
- Select the best-performing model.

Suggested algorithms:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost (optional)
- Support Vector Machine
- K-Nearest Neighbours
- Linear Regression

Part 4 — Model Evaluation & Optimisation (30 Marks)

Evaluate every model using suitable evaluation metrics.

Examples include:

Classification:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Confusion Matrix

Regression:

- MAE
- MSE
- RMSE
- R^2 Score

Improve model performance by:

- Hyperparameter tuning (GridSearchCV or RandomizedSearchCV)
- Cross-validation
- Feature selection
- Handling class imbalance where applicable

Explain why the final model was selected.

Part 5 — Model Interpretation & Explainability (20 Marks)

Interpret your best-performing models.

Include:

- Feature Importance
- Permutation Importance (where applicable)
- SHAP or LIME explanations (if applicable)
- Error analysis
- Model limitations
- Business interpretation of predictions

Explain how your model supports business decision-making.

Part 6 — Machine Learning Deployment (20 Marks)

Develop a simple application that demonstrates your best-performing model.

Examples include:

- Streamlit Web Application
- Gradio Interface
- Flask API
- FastAPI Endpoint

The application should:

- Accept user input.
- Generate predictions.
- Display prediction confidence (where appropriate).
- Present results in a user-friendly interface.

Part 7 — Business Insights & Recommendations (20 Marks)

Based on your models:

- Identify the most important business drivers.
- Recommend operational improvements.
- Explain how management can use predictive insights.
- Discuss model limitations.
- Suggest future AI opportunities.
- Provide **at least ten strategic recommendations** supported by your findings.

Part 8 — Documentation & Presentation (20 Marks)

Prepare:

Technical Report

Include:

- Problem Definition
- Data Preparation
- Feature Engineering
- Model Development
- Model Evaluation
- Model Interpretation
- Business Insights
- Recommendations
- Conclusion

Executive Presentation

Prepare a **10–15 slide presentation** summarising:

- Business Problem
- Machine Learning Approach
- Model Performance
- Key Findings
- Business Recommendations
- Future AI Roadmap

Expected Deliverables

```
Week5_MachineLearning_SwiftLog_YourName/

├── Data/
│   ├── Original_Datasets/
│   ├── Processed_Datasets/
│   └── Feature_Engineered_Dataset.csv
│
├── Notebook/
│   └── Week5_MachineLearning.ipynb
│
├── Models/
│   ├── Best_Model.pkl
│   ├── Model_Comparison.xlsx
│   └── Feature_Importance.png
│
├── App/
│   ├── app.py
│   ├── requirements.txt
│   └── assets/
│
```

```
├── Reports/
│   ├── Technical_Report.pdf
│   └── Executive_Summary.pdf
│
├── Presentation/
│   └── Executive_Presentation.pptx
│
├── README.md
└── References/
```

PART 9 — ASSESSMENT CRITERIA

Section	Marks
Data Preparation & Feature Engineering	20
Exploratory Data Analysis	20
Predictive Model Development	50
Model Evaluation & Optimization	30
Model Interpretation & Explainability	20
Machine Learning Deployment	20
Business Insights & Recommendations	20
Technical Report & Presentation	20
Documentation & Project Organization	20
Total	**220 Marks**

Generative AI Engineering Track

Week 5 – Logistics Generative AI

Mission GenAI05: Building AI-Powered Logistics Assistants & Intelligent Supply Chain Solutions

Client

SwiftLog Logistics Limited

SwiftLog Logistics Limited is a rapidly expanding logistics and supply chain company responsible for managing customer orders, inventory, suppliers, warehouses, deliveries, shipments, employees, and product returns across multiple regions.

As business operations continue to grow, management receives thousands of operational reports, customer enquiries, delivery complaints, supplier communications, and inventory updates every week. Decision-makers spend significant time searching through operational data before taking action.

To improve efficiency and decision-making, SwiftLog wants to leverage **Generative Artificial Intelligence** to automate reporting, provide intelligent business assistants, summarize logistics information, answer operational questions, generate insights from company data, and improve customer support.

As a **Junior Generative AI Engineer** at **AnalystLab Africa Consulting**, your responsibility is to design, build, and evaluate AI-powered solutions that combine Large Language Models (LLMs) with SwiftLog's logistics datasets to improve operational efficiency, customer experience, and executive decision-making.

Business Objectives

Your AI solutions should help SwiftLog:

- Improve customer support.
- Generate automated business reports.
- Summarize logistics operations.
- Answer questions from operational datasets.
- Improve executive decision making.
- Automate repetitive logistics tasks.
- Enhance employee productivity.
- Lay the foundation for enterprise AI adoption.

Datasets to Use

Use the following datasets to build your AI-powered solutions.

Dataset	Purpose
SwiftLog_Customers.csv	Customer information for personalised AI responses and customer support.
SwiftLog_Orders.csv	Order status, order history and order-related question answering.
SwiftLog_Order_Items.csv	Product-level transaction information for AI recommendations and summaries.
SwiftLog_Products.csv	Product catalogue used for intelligent product search and recommendations.
SwiftLog_Suppliers.csv	Supplier information for supplier support and reporting assistants.
SwiftLog_Warehouses.csv	Warehouse information for operational assistance and reporting.

Dataset	Purpose
SwiftLog_Inventory.csv	Inventory data for inventory assistants and stock summaries.
SwiftLog_Shipments.csv	Shipment tracking and logistics monitoring.
SwiftLog_Delivery_Performance.csv	Delivery performance analysis and delay explanations.
SwiftLog_Employees.csv	Internal employee information for operational assistants.
SwiftLog_Returns.csv	Product return information for customer service automation.
SwiftLog_Data_Dictionary.xlsx	Reference guide for understanding business terminology and dataset fields.

Dataset Integration Requirements

Your AI solutions should combine multiple datasets to answer business questions effectively.

Examples include:

AI Solution	Recommended Datasets
Customer Support Assistant	Customers + Orders + Returns
Order Tracking Assistant	Orders + Shipments + Delivery_Performance
Inventory Assistant	Inventory + Products + Warehouses
Executive Reporting Assistant	All datasets
Supplier Assistant	Suppliers + Shipments + Delivery_Performance
Product Recommendation Assistant	Products + Orders + Order_Items

Learning Objectives

By completing this project, you should be able to:

- Design effective prompts for business applications.
- Use Large Language Models (LLMs) to solve real-world problems.
- Build Retrieval-Augmented Generation (RAG) applications.
- Create AI assistants using structured business data.
- Evaluate AI-generated responses.
- Implement AI guardrails for safe and reliable outputs.

- Develop multi-step AI workflows and agents.
- Build simple AI-powered web applications.
- Communicate AI solutions to technical and business stakeholders.

Project Tasks

Part 1 — Data Preparation for Generative AI (15 Marks)

Prepare the logistics datasets for use with LLM applications.

Your tasks should include:

- Explore all provided datasets.
- Clean text-based fields where necessary.
- Merge relevant datasets for AI applications.
- Create structured business knowledge documents from the datasets.
- Generate summaries of each dataset that can be used as AI context.
- Document your data preparation workflow.

Part 2 — Prompt Engineering (20 Marks)

Design **at least 20 high-quality prompts** for different logistics scenarios.

Your prompts should cover:

- Customer support
- Delivery tracking
- Inventory management
- Supplier management
- Executive reporting
- Product recommendations
- Return handling
- Operational troubleshooting
- Data analysis
- Business strategy

For each prompt:

- State the objective.
- Write the prompt.
- Evaluate the response.
- Improve the prompt using prompt engineering techniques (e.g., zero-shot, few-shot, role prompting, chain-of-thought where appropriate, structured output prompts).

Part 3 — Retrieval-Augmented Generation (RAG) Solution (30 Marks)

Develop a RAG application using the logistics datasets.

Your solution should:

- Convert relevant documents into embeddings.
- Store embeddings in a vector database (e.g., ChromaDB or FAISS).
- Retrieve relevant information based on user queries.
- Generate accurate, context-aware responses grounded in the provided data.
- Include citations or references to retrieved information where appropriate.

Evaluate the RAG system using at least 10 sample business questions.

Part 4 — AI-Powered Business Assistant (35 Marks)

Develop an AI assistant capable of answering logistics-related questions.

The assistant should support tasks such as:

- Checking order status.
- Summarising delivery performance.
- Identifying delayed shipments.
- Recommending inventory replenishment.
- Explaining product returns.
- Summarising supplier performance.
- Answering executive business questions.
- Generating operational reports.

The assistant should include:

- Conversation memory (where appropriate).
- Error handling.
- Structured responses.
- User-friendly interface (Streamlit, Gradio, or similar).

Part 5 — AI Workflow Automation (25 Marks)

Build an AI workflow that automates at least **three** business processes.

Examples include:

- Daily executive report generation.
- Automatic shipment delay summaries.
- Inventory restocking recommendations.
- Weekly supplier performance reports.

- Customer complaint summarisation.
- Return reason categorisation.
- Email draft generation for logistics updates.

Document the workflow and explain the business value.

Part 6 — AI Evaluation & Responsible AI (20 Marks)

Evaluate your AI solution by assessing:

- Accuracy
- Relevance
- Hallucination risk
- Consistency
- Response quality
- Bias and fairness considerations
- Safety and guardrails

Propose improvements based on your evaluation.

Part 7 — Business Insights & Recommendations (20 Marks)

Based on your AI solution:

- Identify opportunities for automation.
- Explain how AI improves operational efficiency.
- Recommend additional AI use cases for SwiftLog.
- Discuss implementation challenges.
- Suggest governance and data privacy considerations.
- Provide **at least ten strategic recommendations** for adopting Generative AI across the organisation.

Part 8 — Documentation & Presentation (15 Marks)

Prepare a professional submission including:

Technical Report

Include:

- Problem Definition
- AI Solution Architecture
- Data Preparation
- Prompt Engineering Strategy

- RAG Design
- AI Assistant Development
- Evaluation Results
- Business Insights
- Recommendations
- Conclusion

Executive Presentation

Prepare a **10–15 slide presentation** covering:

- Business Problem
- AI Solution
- Live Demonstration (screenshots or walkthrough)
- Key Results
- Business Benefits
- Future AI Roadmap

Expected Deliverables

```
Week5_GenerativeAI_SwiftLog_YourName/

├── Data/
│   ├── Original_Datasets/
│   ├── Processed_Knowledge_Base/
│   └── Embeddings/
│
├── Notebook/
│   └── Week5_GenerativeAI.ipynb
│
├── Application/
│   ├── app.py
│   ├── prompts/
│   ├── knowledge_base/
│   ├── vector_store/
│   └── requirements.txt
│
├── Reports/
│   ├── Technical_Report.pdf
│   └── AI_Evaluation_Report.pdf
│
├── Presentation/
│   └── Executive_Presentation.pptx
│
├── README.md
└── References/
```

PART 9 — ASSESSMENT CRITERIA

Section	Marks
Data Preparation for AI	15
Prompt Engineering	20
RAG Development	30
AI Business Assistant	35
AI Workflow Automation	25
AI Evaluation & Responsible AI	20
Business Insights & Recommendations	20
Technical Report & Presentation	15
Documentation & Project Organisation	20
Total	**200 Marks**

Expected Learning Outcomes

By the end of this project, interns should be able to:

- Prepare structured business data for use with Large Language Models.
- Design and evaluate effective prompts for enterprise logistics scenarios.
- Build Retrieval-Augmented Generation (RAG) applications using real business datasets.
- Develop AI-powered assistants that answer logistics and operational questions.
- Automate business workflows using Generative AI.
- Evaluate AI systems for quality, safety, and reliability.
- Present AI solutions that deliver measurable business value in a real-world logistics consulting engagement.

Project Management Track

Week 5 – Logistics Project Management

Mission PM05: Planning and Managing a Supply Chain Analytics & AI Transformation Project

Client

SwiftLog Logistics Limited

SwiftLog Logistics Limited is a rapidly growing logistics and supply chain company responsible for managing customer orders, inventory, suppliers, warehouses, deliveries, shipments, employees, and product returns across multiple regions.

As the business expands, SwiftLog has initiated a strategic digital transformation programme aimed at improving operational efficiency through **Data Analytics, Data Science, Machine Learning, and**

Generative AI. The programme involves multiple technical teams working together to deliver intelligent logistics solutions.

As a **Junior Project Manager** at **AnalystLab Africa Consulting**, your responsibility is to successfully plan, coordinate, monitor, and manage the execution of the SwiftLog Analytics & AI Transformation Project. Your role is to ensure that project objectives are delivered on time, within scope, within budget, and according to quality standards while effectively managing project risks, stakeholders, resources, communications, and deliverables.

Business Objectives

Your project management plan should help SwiftLog:

- Successfully deliver the Analytics & AI Transformation Project.
- Improve collaboration between technical teams.
- Ensure timely project delivery.
- Minimise project risks.
- Improve communication among stakeholders.
- Manage project scope effectively.
- Monitor project performance.
- Deliver measurable business value.

Datasets to Use

Although Project Managers are not expected to perform technical data analysis, they should understand the datasets being used by the technical teams to effectively plan project activities, estimate resources, identify dependencies, and coordinate deliverables.

Dataset	Purpose
SwiftLog_Customers.csv	Understand customer-related project requirements and stakeholder expectations.
SwiftLog_Orders.csv	Understand order processing workflows and business processes.
SwiftLog_Order_Items.csv	Understand product transaction workflows and reporting requirements.
SwiftLog_Products.csv	Support project planning for product analytics deliverables.
SwiftLog_Suppliers.csv	Understand supplier management requirements and external stakeholder coordination.

Dataset	Purpose
SwiftLog_Warehouses.csv	Support planning for warehouse analytics and operational improvements.
SwiftLog_Inventory.csv	Understand inventory management processes to define project scope.
SwiftLog_Shipments.csv	Support logistics workflow mapping and operational planning.
SwiftLog_Delivery_Performance.csv	Understand delivery performance metrics and project success criteria.
SwiftLog_Employees.csv	Assist with resource planning and organisational analysis.
SwiftLog_Returns.csv	Understand return management processes and improvement opportunities.
SwiftLog_Data_Dictionary.xlsx	Reference document for understanding business terminology and project requirements.

Project Dependencies

Your project plan should consider the dependencies between the learning tracks.

Learning Track	Primary Responsibility
Data Analytics	Business Intelligence dashboards and KPI reporting
Data Science	Data preparation, statistical analysis and feature engineering
Machine Learning	Predictive model development and evaluation
Generative AI	AI assistants, RAG solutions and workflow automation
Project Management	Planning, coordination, monitoring and governance

Learning Objectives

By completing this project, you should be able to:

- Develop a comprehensive project charter.
- Define project scope and objectives.
- Develop a Work Breakdown Structure (WBS).

- Create a realistic project schedule.
- Allocate resources effectively.
- Identify and manage project risks.
- Develop stakeholder engagement strategies.
- Create project communication plans.
- Monitor project performance using project management tools.
- Prepare professional project documentation.

Project Tasks

Part 1 — Project Initiation (20 Marks)

Develop a **Project Charter** that includes:

- Project title
- Business case
- Project objectives
- Project scope
- Deliverables
- Success criteria
- Assumptions
- Constraints
- High-level risks
- Project sponsor
- Project manager
- Key stakeholders

Part 2 — Project Planning (35 Marks)

Develop a comprehensive Project Management Plan including:

- Scope Statement
- Work Breakdown Structure (WBS)
- Milestone Plan
- Project Schedule (Gantt Chart)
- Resource Plan
- Budget Estimate
- Procurement Plan
- Quality Management Plan
- Communication Management Plan
- Change Management Plan

Use Microsoft Project, Excel, ClickUp, Trello, Jira, Asana, Notion, or another recognised project management tool.

Part 3 — Stakeholder & Resource Management (20 Marks)

Develop:

- Stakeholder Register
- Stakeholder Analysis Matrix
- Power-Interest Grid
- RACI Matrix
- Resource Allocation Matrix
- Team Responsibility Plan
- Communication Schedule

Explain how each stakeholder will be engaged throughout the project lifecycle.

Part 4 — Risk, Issue & Change Management (30 Marks)

Develop:

- Risk Register
- Risk Assessment Matrix
- Issue Log
- Assumption Log
- Dependency Log
- Change Request Form
- Change Log
- Risk Response Plan
- Contingency Plan

Identify at least **10 project risks** and propose appropriate mitigation strategies.

Part 5 — Project Monitoring & Control (30 Marks)

Design a project monitoring system that includes:

- Project Status Dashboard
- Milestone Tracking
- Budget Tracking
- Schedule Performance Indicators
- Resource Utilisation Dashboard
- Project KPI Dashboard
- Variance Analysis
- Progress Reporting Template
- Lessons Learned Register
- Decision Log

Explain how project performance will be monitored throughout execution.

Part 6 — Agile Project Management (20 Marks)

Plan the project using Agile principles.

Develop:

- Product Backlog
- Sprint Backlog
- Sprint Plan
- Sprint Review Template
- Sprint Retrospective Template
- User Stories
- Acceptance Criteria
- Kanban Board or Scrum Board

Explain why Agile is appropriate for this Analytics & AI transformation project.

Part 7 — Project Governance & Quality Assurance (20 Marks)

Develop:

- Governance Structure
- Decision-Making Framework
- Quality Assurance Plan
- Quality Control Checklist
- Document Approval Workflow
- Escalation Process
- Meeting Schedule
- Project Closure Checklist

Explain how governance ensures successful project delivery.

Part 8 — Executive Reporting & Presentation (25 Marks)

Prepare professional project documentation including:

Project Management Report

Include:

- Executive Summary
- Project Charter
- Scope
- Schedule

- Budget
- Risks
- Stakeholder Plan
- Governance
- Monitoring Strategy
- Recommendations
- Conclusion

Executive Presentation

Prepare a **10–15 slide presentation** covering:

- Business Problem
- Project Objectives
- Project Plan
- Timeline
- Risk Management
- Governance
- Expected Outcomes
- Recommendations

Expected Deliverables

```
Week5_ProjectManagement_SwiftLog_YourName/

├── Project_Charter/
│   └── Project_Charter.pdf
│
├── Planning/
│   ├── WBS.xlsx
│   ├── Gantt_Chart.xlsx
│   ├── Resource_Plan.xlsx
│   ├── Budget.xlsx
│   └── Communication_Plan.pdf
│
├── Governance/
│   ├── Risk_Register.xlsx
│   ├── Issue_Log.xlsx
│   ├── Change_Log.xlsx
│   ├── Stakeholder_Register.xlsx
│   ├── RACI_Matrix.xlsx
│   └── Quality_Plan.pdf
│
├── Agile/
│   ├── Product_Backlog.xlsx
```

Part 12 - ASSESSMENT CRITERIA

Section	Marks
Project Initiation	20
Project Planning	35
Stakeholder & Resource Management	20
Risk, Issue & Change Management	30
Project Monitoring & Control	30
Agile Project Management	20
Project Governance & Quality Assurance	20
Executive Report & Presentation	25
Documentation & Project Organisation	20
Total	**220 Marks**

Professional Development Requirement

As part of building your professional portfolio:

- Update your GitHub repository with your Week 5 project.
- Publish a LinkedIn post describing your project
- Publish an X (Twitter) post summarizing your project.
- Tag AnalystLab Africa on LinkedIn.
- Tag @analystlabafric on X.
- Include the hashtag #AnalystLabAfrica on both platforms.

Professional Branding Reminder

AnalystLab Africa encourages every intern to document their learning journey professionally on LinkedIn and X. Consistently sharing your work helps build your portfolio and increases your visibility to recruiters and employers.